

Chapter

Perspective Chapter: Novel Slow-Release Nanocomposite Fertilizers

Atena Mirbolook

Abstract

Slow-release fertilizers (SRF) create a physical barrier and prevent the rapid release of nutrients. These fertilizers are produced in two ways: coated and matrix. SRFs can reduce nutrient loss such as nitrogen and cause gradual use according to the plant's needs. This will increase productivity and make fertilizer more effective. The process of producing fertilizers is increasing every year in the world. It can also be predicted that this trend will continue in the coming years. Nanotechnology-based fertilizers also are a new strategy to enhance agricultural yields and show great potential as viable options in the fertilizer industry. Recently slow- and controlled-release nano-fertilizers (SRNF and CRNF) have advanced through the improvement of nanocomposites or coating techniques with the help of various chemical things. SRNFs are more effective than usual nano-fertilizers because these deliver elements in a controlled method and can be adjusted by many environmental and physical motivations (such as pH, temperature, humidity). The application of controlled-release fertilizers reduces nutrient loss due to evaporation and leaching and provides a carefully designed nutrient-release system that is aligned with the goal of sustainable agriculture.

Keywords: nano-fertilizers, slow-release, nutrient, nanocomposite, preparation

1. Introduction

Agriculture in the world is in crisis due to population increase, high demand for quality food, and adverse effects of climatic, biotic, and abiotic stresses on global food production [1]. Predictions show that the world's population will reach 9.7 billion by 2050, requiring food security with naturally functioning water, energy, and non-renewable feedstocks, productive land, phosphate reserves, and ecosystems. Given the current alarms about food safety in the future, the design and application of sustainable solutions are urgently needed. With the use of current resources and technologies, agricultural production is now at its maximum. Revolutionary nanotechnology in the energy, agriculture, and health industries can revolutionize the approach to agriculture in the world. The productivity of agriculture is decreasing day by day

because this particular sector is facing a lot of problems, among them the reduction of fertile land, mismanagement in the use of fertilizers, lack of suitable food sources, and reduction of soil health and organic matter [2]. Since the development of fertile agricultural land is not possible, proper management of fertilizers, disease, and stress (biotic and abiotic) may be an ideal way to increase crop production [3]. For many years, various types of chemical fertilizers have been used to achieve the goals of farmers and high yield.

Recently, the science of nanotechnology has emerged as an attractive topic for major changes in the agricultural sector. Nano compounds affect various aspects of agriculture such as seed germination, plant growth and development [4], enrichment of fertilizers [3], plant diseases [5], and plant stress management [6]. The small size of these particles (usually 1–100 nm) and their high specific surface area led to their increased absorption and efficiency in systems. By using these compounds, we can predict higher fertilization efficiency, increased plant productivity, and minimum costs for farmers [7]. However, nano-fertilizers cannot solve all the problems related to fertilization, because some problems such as leaching and accessibility and transfer of nutrients still exist. In addition, improving nutrient use efficiency (NUE) is a critical development area.

2. Slow- and controlled-release nano-fertilizers

Slow- and controlled-release nano-fertilizers (SRNFs and CRNFs) are a step ahead of conventional nanoparticles (iron, copper, zinc, manganese oxide nanoparticles, etc.) that can solve the problems of nutrient absorption and transport more effectively. These include coated nano-fertilizers (CRNF) or composite nano-fertilizers (SRNF) that release nutrients in a controlled manner, thereby reducing the possibility of leaching and wastage of nutrients in the soil [8]. SRN are usually nanocomposite fertilizers with a complex polymer network, which are mainly used to prepare controlled-release fertilizers (CRFs). They use coated nano-fertilizers to prepare SRFs. These fertilizers retain nutrients and due to having external coatings or composition in a complex matrix, the steps of releasing nutrients from their structure happen slowly or in a controlled manner. Synthesis of CRFs is achieved by various encapsulation and coating methods, while SRFs can be prepared using a variety of composite materials including hydroxyapatite, hydrogel, chitosan, alginate, etc. [9]. In SRF, nutrient release can be stimulated or controlled through various parameters such as pH, temperature, and soil moisture, and as a result, regulation of nutrient release is possible [10, 11]. Additionally, the release of a specific element can be tracked or monitored by various analytical techniques (such as AAS, ICP-AES, ICP-MS, airborne, electrochemical, and mobile sensors).

Regarding the use of nanotechnology, attention should also be paid to environmental pollution. This pollution happens with the release of these nanoparticles in the environment [12, 13].

Nutrient elements in the structure of nano-fertilizers can be transferred to plants in the following ways:

- A. Targeted delivery of nutritional elements on a nanoscale in the form of particles or emulsions.

- B. Controlled release of nano-fertilizers by encapsulating nutrients to achieve the goal in a controlled manner, which includes three types:
- Slow-release systems (releasing the nutrients from the coating or matrix polymer for a long period).
 - Rapid-release systems (the nutrients are released immediately from the target surface).
 - Conditional release systems (the release of nutrients is performed based on external factors such as pH, temperature, humidity, or magnetic/ultrasonic pulse).
- C. Special materials are included in the matrix of organic biochemical polymers that act as carriers [14].

Among these nano-fertilizers, SRNFs are currently considered a suitable instrument for the synthesis of nano-fertilizers of macronutrient and micronutrient essential elements. CRNF and SRNF are often used interchangeably [15]. But they are different and one of the differentiating factors between CRNF and SRNF is the existence of cover. SRNFs are nanocomposites in which nutrients are combined into a compound with high molecular weight and a complex structure, and these are released through the diffusion of soluble and degradable compounds in water and microbial decomposition [16]. SRNF release nutrients more slowly than usual nanoparticles, but they cannot handle the release of nutrients [10, 17].

Slow-release fertilizers (SRFs), distinct CRFs, are not encapsulated (**Figure 1**). The most common SRFs, provide nitrogen (N) at a slower rate than when a readily soluble nitrogen source (such as ammonium sulfate, ammonium nitrate, or urea) is used. In one such fertilizer, fertilizer producers synthesized a SRF by chemical reaction of a nitrogen source molecule with an aldehyde, for example, urea-formaldehyde or methyl urea. Delayed release of nitrogen is done by microbial action in the growth medium. This process is done in such a way that the long chain molecules are slowly broken and finally nitrogen is released in the form of ammonium and will be converted into nitrate and absorbed.

It should be highlighted that the release time in SRF cannot be controlled since the efficiency of microbial organisms in molecular decomposition depends on other

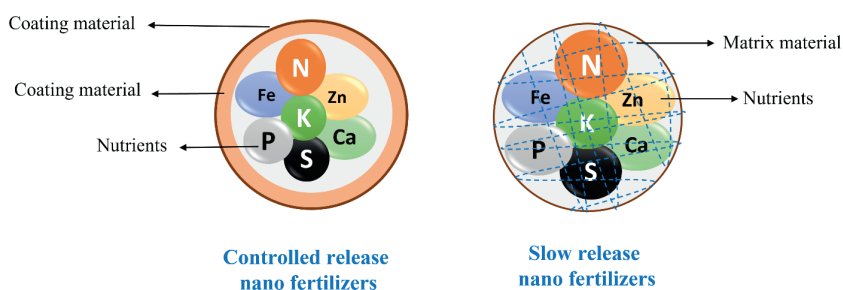


Figure 1.
 Controlled and slow-release nano-fertilizers.

factors such as the growth environment, humidity, and temperature. Also, release time cannot be expected for more than 2 or 3 months.

CRFs differ fundamentally from SRFs both in technology and in the way, nutrients are released. Soluble necessary plant nutrients are encapsulated in an organic resin or polymer coating to form coated granules. This coating results in the delayed release of nutrients in a SRF.

In an SRF fertilizer, the coating on the granules acts as a selectively permeable or semipermeable membrane, which is a barrier to some molecules but allows others to pass through. When a CRF is placed in a suitable moisture condition, water passes one way from the coating into the fertilizer granule, and this phenomenon is osmosis. The absorbed water moderately dissolves the nutrients inside the coated fertilizer granule to make a highly concentrated solution. Then the hydrostatic pressure inside the capsule increases, and when the hydrostatic pressure on both sides of the capsule is equal, water will no longer enter (**Figure 2**).

The process of movement of nutrients toward the plant occurs as a diffusion phenomenon, that is, the movement of molecules from a liquid with a higher concentration to a liquid with a lower concentration. The reason for this phenomenon is related to the coating structure, which has small micropores. When plants are watered, the hydrostatic pressures inside and outside the capsule become unequal, and small amounts of dissolved nutrients are released into the growing medium by diffusion through these pores.

The type of coating agent determines the rate and pattern of nutrient release. CRNFs are known as coated fertilizers wherever the factors determining the rate, duration, and pattern of nutrient release can be fully controlled during preparation [16–18]. The coating and matrix materials used to prepare CRNF or SRNF can be inorganic or organic. Minerals include phosphogypsum, bentonite, sulfur, etc. Organic polymers can be synthetic (polyethylene, polyurethane, alkyd resin, etc.) or natural (chitosan, starch, cellulose, etc.) [10, 15].

SRNFs have properties that are very useful for agricultural activity. They are one of the most effective ways to deliver nutrients to plants because they can be designed to release nutrients slowly and deliver them to plants with minimal wastage.

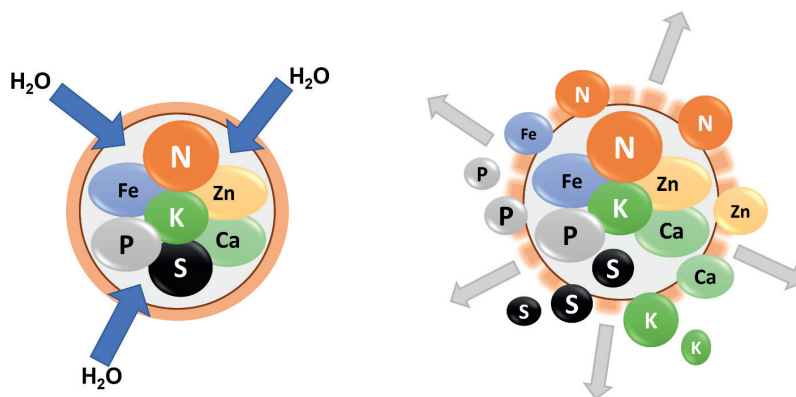


Figure 2. Diffusion mechanism for releasing nutrients from coated granules of fertilizers, nutrients moved out through the coated materials.

3. Types of nanocomposites-based SRNFs

3.1 Hydroxyapatite-based nanocomposites

Hydroxyapatite (HA) nanoparticles are a group of minerals that belong to calcium phosphate and have a chemical composition $(\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2)$ that is very similar to the mineral found in human bones and teeth [19]. These nanoparticles have been extensively researched due to their importance in materials science, biology, and pharmaceuticals [20]. According to research, hydroxyapatite nanoparticles (HANPs) are between 40 and 60 nm in size. They can also be easily activated with various biomolecules such as proteins and peptides to enhance their biological activity. The HA nanoparticles are biocompatible and nontoxic, making them promising medical and agricultural materials [20, 21]. When suspended in water, they exhibit colloidal properties and can carry various other nutrients. Gao et al. [22] showed that HANPs in combination with urea perform a dual function, that is, in addition to releasing sufficient nitrogen and phosphorus, they can also reduce the rate of urea decomposition in the soil and as a result, improve agricultural efficiency [22]. A recent review article showed that nano-hydroxyapatite particles can have both adverse and beneficial effects on plant growth and improvement, and recommended that more research is needed on different plants [23]. The desired food nutrients can be mixed with HANPs to synthesize hydroxyapatite nanocomposites and use a slow-release process.

3.2 Nanocomposites based on hydrogel

Hydrogel is a three-dimensional network of hydrophilic polymers that can swell in water and retain a significant amount of water in its structure. Due to their significant water content, hydrogels exhibit remarkable flexibility, which is very similar to natural tissues. It facilitates water retention in their structure and physiochemical cross-linking between separate polymer chains in the hydrogel. There are different hydrophilic groups including $-\text{NH}_2$, $-\text{CONH}_2$, $-\text{COOH}$, $-\text{OH}$, $-\text{SO}_3\text{H}$, and $-\text{CONH}$ in the hydrogel [24].

Hydrogels have recently attracted considerable attention due to their potential participation in agricultural activities and can be used as nanocomposite compounds for the synthesis of SRNFs [25, 26]. Noppakundilongrat et al. [27] prepared a novel three-layer coated NPK hydrogel fertilizer using a coating technique [27]. In this synthesis method, NPK bits were immersed in the solution of polyvinyl alcohol and chitosan. Cross-linking between cellulose layers was done using glutaraldehyde. The results show that the cross-linking of the chitosan layer increases the water infiltration and the coating layer increases the water suspension time.

3.3 SRNFs based on chitosan

Chitosan, a natural polysaccharide composed of N-acetyl-D-glucosamine and D-glucosamine groups, is a biodegradable polymer that fits fine into SRNF formulations [28]. This acetylated chitin absorbed maximum water due to its super absorbent properties. In addition, their biocompatible, cost-effective, hydrophilic, non-toxic, and biodegradable nature makes them suitable for agricultural use [29, 30]. This biopolymer has been widely studied due to its swelling ability and nutrient-release capacity [30, 31]. Chitosan can play an important role with cross-linking agents, such

as sodium tripolyphosphate, epichlorohydrin, glutaraldehyde, etc., to entrap nutrients [29]. Chitosan nanoparticles can be simply mixed with nutrients to control the release of that by forming SRNF [32].

Araujo et al. [33] formulated chitosan-based coating materials using humic substances (peat, humic acid, humin) [33]. It has been reported that depending on the type of humic substances and the pH of the environment, according to the functional group and possible interactions of each compound with urea, it will have a different effect on the release rate. Huey et al. [34] included the use of allicin, a urease inhibitor, in chitosan/starch composites [34]. Allicin has been reported to reduce the rate of urea hydrolysis and delay the availability of nutrients to plants.

3.4 Graphene oxide or carbon-based nanocomposites

Graphene oxide (GO) is a single atomic layer material obtained through the oxidation of graphite. Numerous studies show that GO is a non-toxic and biocompatible material with a large surface area and a large number of reactive oxygen functional groups [35]. Due to their high specific surface, materials containing graphene provide a suitable substrate for loading nutrients and preparing SRFs [36]. Haydar et al. study [4] showed that iron and manganese nanoparticles were integrated into graphene using a doping technique, which led to the preparation of graphene-based nanocomposites with micronutrients [4].

3.5 SRNFs based on diatomite

Diatomite (D) is a type of sedimentary rock that includes a shell of hard silicified diatomite and some other minerals such as clay minerals, feldspar, etc. Diatomite has unique physical and chemical properties including high porosity, high adsorption capacity, resistance, small particle size, and a high specific surface area of $10\text{--}30\text{ m}^2\text{ g}^{-1}$. Diatomites are compounds that have a high cation exchange capacity, and these characteristics make them important for use in the structure of SRFs [37]. Hydroxyl groups on the surface of diatomite are the main reaction sites. In addition, acidic sites are also seen on the surface of diatomite, which are suitable for surface reactions [38]. Nano-clay minerals have higher surface area and charge density than micro-sized clay minerals [39]. Yuvaraj and Subramanian [40] used a nano zeolite as a zinc carrier and observed that 1176 hours were spent to release zinc from the nano zeolite structure [40]. Also, in another research, it was found that nano bentonite can retain cadmium, chromium, and copper elements up to 74, 99.03, and 99.18% more than normal ionic solution [41]. Meanwhile, the adsorption efficiency in micro bentonite for cadmium and chromium was reported as 82.4% and 55%, respectively [42]. In research conducted by Mirbolook et al. [43], it was determined that the binding of urea-zinc complex on the porous surface of diatomite and nano diatomite can reduce the release rate of urea in the water environment and nitrogen and zinc in the soil environment [43]. The newly synthesized fertilizer was known as SRF and showed high efficiency in maintaining the mineral forms of nitrogen and zinc in the soil. The results showed that the diffusion of urea in water follows a sigmoidal pattern. In the case where urea was present in the matrix with nano-diatomite, about 20% of urea was released in water after 12 hours, while in normal urea, 40–50% of urea was released after 12 hours. These SRNFs can be considered as a potential material for the slow release of elements in the soil as well as minimizing the cost and environmental problems.

3.6 SRNFs based on carbon dot

Carbon nanomaterials are one of the environmentally friendly components and are widely used for agricultural development [44]. Carbon dots (CDs) [45], carbon nanotubes [46], and fullerenes [47] have all been examined for plants, but CDs have attracted specific attention due to their small size (less than 10 nm). The CDs have numerous advantages, including simplicity of synthesis and surface modification, high water solubility, fluorescence properties, low toxicity, and biocompatibility [48]. Carbon dots are simply uptake by plants and can promote growth. These components can increase photosynthesis, nutrient uptake, and nitrogen fixation. Studies have also shown that CDs have a positive effect on plant growth and resistance to environmental stresses [49]. In the study of Alikhani et al. [50], a carbon dot was used as a new carrier for the slow release of zinc elements. For this purpose, Zn-doped carbon dot (Zn-NCDs) was synthesized by a simple hydrothermal method [50]. Then, the effect of zinc-NCDs as a source of zinc on wheat plants under greenhouse conditions was investigated. The results showed that the application of slow release through carbon nanocarrier significantly increased wheat growth indices, biomass, grain yield, nitrogen, and zinc content. In addition, the molar ratio of phytic acid to zinc in wheat grains decreased significantly, which affects the bioavailability of zinc in wheat grains. Finally, although significant progress has been made in the field of the effect of CDs on plants, more research is still needed for the application of CDs in the preparation of SRNFs in agriculture.

3.7 SRNFs based on alginate

One of the natural polymers used as a coating in fertilizer granules is sodium alginate. Carbohydrate alginate derived from brown algae is a natural polymeric polysaccharide that consists of two monosaccharides [51]. These two are β -D-mannuronic acid and α -L-guluronic acid. These two monomers are connected in a straight line to form a macromolecular structure of alginate [51]. Sodium alginate can react with polyvalent cations and form a gel, and the gelation process is thermally irreversible [52]. In terms of biodegradability, compatibility, and adhesion, it shows good performance, which is suitable for controlling the release rate of food elements. The study by Meng et al. used a combination of poly (vinyl alcohol) (PVA)/sodium alginate (SA) polymer membranes as environmentally friendly and biodegradable coatings for SRFs [53]. The use of these compounds greatly reduced the release of nitrogen from the fertilizer structure. Chen et al. [54] worked on the fabrication of potassium chloride encapsulated in a starch alginate matrix, crosslinked by calcium chloride to form a hydrogel bead [54]. The result shows that a higher concentration of SA reduces

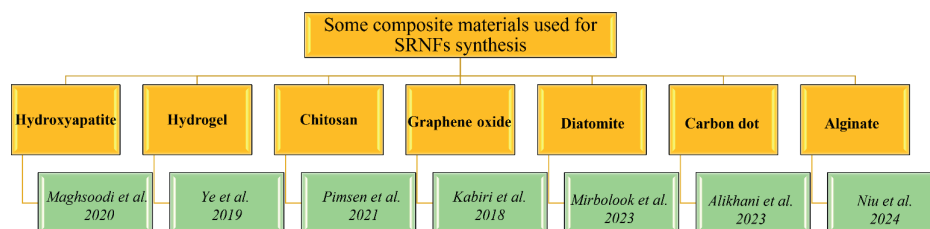


Figure 3.
 Various types of composite material used to prepare slow-release nano-fertilizers.

the swelling of fertilizer granules and thus reduces the release of potassium. In other words, with higher SA concentration, the emission intensity was better controlled. In contrast, it was also observed that the dissolution of the sample in water increased after 24 hours with increasing SA content. The importance of this research depends on its potential applications in agricultural practices, especially in drought fields. By fine-tuning the ratios of starch, alginate, and potassium, this study opens the way for further improvements in sustainable nutrient delivery systems.

Figure 3 shows a classification of composite material used to prepare slow-release nano fertilizers.

4. Conclusion and future perspectives

The increase in population and the need for healthy food have led to a significant increase in the amount of fertilizer used in agriculture. Recent research has focused increasingly on the development of nanoparticle-based fertilizer synthesis methods. However, the chance of excessive usage of nano-based fertilizers, their interaction with soil macro and microorganisms, and their impact on human health and ecosystems remain unclear, and their allowable dosage limits should be determined. The use of controlled nano-fertilizers in agricultural systems has many advantages, but the increasing use of their synthesis methods in laboratories and their use in cultivated systems should be further investigated.

Author details

Atena Mirbolook
Faculty of Agriculture, Department of Soil Science, Urmia University, Urmia, Iran

*Address all correspondence to: mirbolook@um.ac.ir

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